Alan Ning

Software Engineer / Problem Solver

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github.com/askldjd (Github Profile)

Work

Software Architect

Manifold.ai

Oct 2021 – Present

Founding engineer of the Manifold.ai platform, responsible for designing and building core systems across the stack.

- Architected platform infrastructure spanning bioinformatics data pipelines, AI security, and scalable cloud networking.
- Built developer tooling that enables LLM agents to iterate safely, accelerating engineering productivity across the team.

Staff Reliability Engineer

Tenable

Oct 2018 – Oct 2021

I create scalability solutions for Tenable.io.

- Designed and implemented a custom Elasticsearch sharding layer to break out a petabytes-scale monolith database into an infinitely scalable number of databases.
- Implemented a caching layer for the Tenable.io edge using Cloudflare Worker. This layer handles over 5000 requests per sec, and saves petabytes of bandwidth per year.

Site Reliability Engineer

Uptime2020 Consultant

Jun 2020 – Sep 2020

Supported various election-related SaaS companies to ensure their product security and reliability during the 2020 US election cycle.

Site Reliability Engineer

United States Digital Service - Department of Veterans Affairs

Sep 2016 – Aug 2018

I began as the SRE for Caseflow and left having architected the VA's migration to AWS.

- Guided the VA cloud migration effort, including Direct Connect configurations, network performance analysis, and VPC design that scales for thousands of tenants.
- Led the SRE team for the Veterans Appeals modernization effort (Caseflow).
- Migrated the 30 year old Oracle database that holds all Veterans Appeals to the cloud using AWS DMS.

Technical Lead

United States Digital Service - Social Security Administration

Nov 2015 – Sep 2016

I began as a React/Node.js coach and left as the Technical Lead on the Disability Case Processing System (DCPS).

- Led the Social Security Disability modernization project, integrating a microservice architecture with legacy mainframe systems.
- Delivered a highly visible MVP under a tight deadline, which led to successful product shipment across many states.

Technical Lead Engineer

Boeing / Digital Receiver Technology

Jun 2005 – Oct 2015

I began as an embedded C++ developer and left as a full stack software developer.

- Wrote high performance C++ code for embedded systems, with a focus on memory safety and concurrency.
- Developed applications across a wide range of platforms, from Android and CUDA to Node/React web apps.
- Grew into a technical lead role across multiple projects, including cybersecurity research.

Education

Johns Hopkins University

Master in **Computer Science**

Sep 2005 – Jun 2007

Rensselaer Polytechnic Institute

Bachelor in **Computer Science / Economics (Summa Cum Laude)**

Sep 2001 – May 2005

Skills

AI & LLM Engineering

Pydantic AI · OpenAI · MCP · Agent Evaluation · Bedrock

Cloud & Infrastructure

AWS · Terraform · Docker · ECS · RDS · Lambda · S3 · Packer

Backend Development

Python · FastAPI · Node.js · SQLAlchemy · PostgreSQL · Snowflake

Data Engineering

Airflow · dbt · Prefect · OpenSearch · Pandas

Observability

OpenTelemetry · Logfire · Structlog

Systems Programming

Modern C++ · C